

PRINCIPLES OF GEOGRAPHIC INFORMATION SYSTEM SEM: VI

SEM VI: UNIT 1



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1. Define GIS. Briefly explain any two capabilities of GIS.**(Apr 19)****Ans:**

GIS (Geographic Information System) is a system designed to capture, store, analyze, manage, and present spatial or geographic data. It integrates various types of data and allows users to visualize, interpret, and understand the relationships, patterns, and trends within the data.

❖ The following four sets of capabilities to handle georeferenced data:**1. Data capture and preparation**

Data capture is tedious job in GIS. A GIS can be used to emphasize the spatial relationships among the objects being mapped. If the data to be used are not already in digital form that is in a form a computer can understand and recognize, various techniques are available to capture the information. Maps can be digitized, or hand traced with a computer mouse, to collect the coordinates of the features. Electronic scanning devices will also convert map lines and points to digits.

In the El Nino case, data capture refers to the collection of sea water temperatures and wind speed measurements. This is achieved by placing buoys with measuring equipment at various places in the ocean. Each buoy measures a number of things: wind speed and direction; air temperature and humidity; and sea water temperature at the surface and at various depths down to 500 metres. For the sake of our example we will focus on sea surface temperature (SST) and wind speed (WS)

2. Data management, including storage and maintenance

This phase requires a decision to be made on how best to represent our data, both in term soft heirspatial properties and the various at tribute values which we need to store. Data manipulation includes data verification, attributes data management, insertion, updating, deleting and retrieval in different forms. .For our example data management refers to the storage and maintenance of the data transmitted by the buoys via satellite communication

3. Data manipulation and analysis

Data analysis can be done, when data has been collected and organized in computer system. In above example, considering data generated at the buoys was processed before map production. A Figure 1.1 reveals that the data being presented are based on the monthly averages for SST and WS (for two months), not on single measurements for a specific date. 1. For each buoy, the average SST for each month was computed, using the daily SST measurements for that month. This is a simple computation. 2. For each buoy, the monthly average SST was taken together with the geographic location, to obtain a georeferenced list of averages, as illustrated in the following table:

Buoy	Geographic position	Dec.1997 avg.SST
B0789	(165°E, 5° N)	28.02°C
B7504	(180°E, 0° N)	27.34°C
B1882	(110°W, 7° 30'S)	25.28°C
...	...	...

4. Data presentation

After the data manipulations our data is prepared for producing output. This data presentation phase deals with putting it all together into a format that communicates the result of data analysis in the best possible way. Before data is presented, we need to consider what the message is that we want to portray, who the audience is, what kind of presentation medium will be used, which rules of aesthetics apply, and what techniques are available for representation.



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2. What is GI System, GI Science and GIS Application? Explain.

(Apr 19)(Apr 23)

Ans:

❖ **GI Science:**

The discipline that deals with all aspects of the handling of spatial data and geoinformation is called geographic information science (often abbreviated to geo-information science or just GI Science).

Geo-Information Science is the scientific field that attempts to integrate different disciplines studying the methods and techniques of handling spatial information.

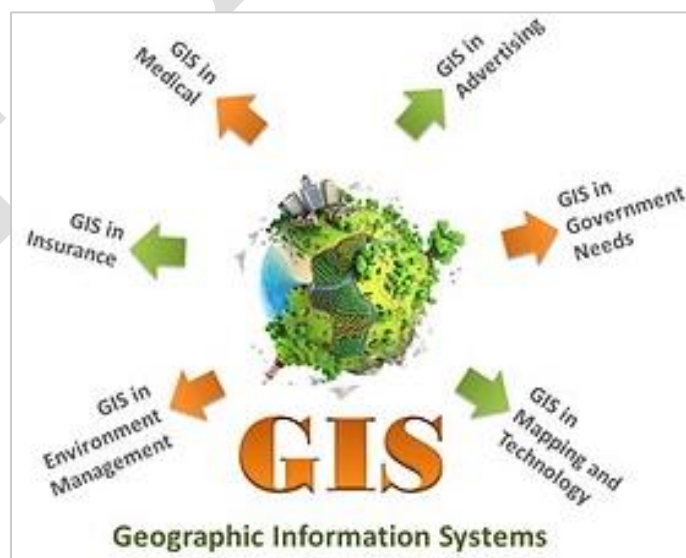
❖ **GI System:**

a geographic information system— in the ‘narrow’ sense—in terms of its functions as is a computerized system that facilitates the phases of data entry, data management, data analysis and data presentation specifically for dealing with georeferenced data.

In the ‘wider’ sense, a functioning GIS requires both hardware and software, and also people such as GI Systems the database creators or administrators, analysts who work with the software, and the users of the end product.

❖ **GIS Application of GIS are as follow:**

1. An urban planner might want to assess the extent of urban fringe growth in her/his city and quantify the population growth that some suburbs are witnessing. S/he might also like to understand why these suburbs are growing and others are not;
2. A biologist might be interested in the impact of slash-and-burn practices on the populations of amphibian species in the forests of a mountain range to obtain a better understanding of long-term threats to those populations;
3. A natural hazard analyst might like to identify the high-risk areas of annual monsoon-related flooding by investigating rainfall patterns and terrain characteristics;



3. How modeling helps in representing real world? Explain.**(Apr 19)****Ans:**

- Modelling' is a term used in many different ways and which has many different meanings. Are presentation of some part of the real world can be considered model because the representation will have certain characteristics in common with the real world. Models as representations come in many different flavours. In the GIS environment, the most familiar model is that of a map. A map is a miniature representation of some part of the real world.
- A 'real world model' is a representation of a number of phenomena that we can observe in reality, usually to enable some type of study, administration, computation and/or simulation. In this book we will use the term application models to refer to models with a specific application, including real-world models and so-called analytical models. The phrase 'data modelling' is the common name for the design effort of structuring a database. This process involves the identification of the kinds of data that the database will store, as well as the relationships between these kinds of data.
- Most maps and databases can be considered static models. At any point in time, they represent a single state of affairs. Usually, developments or changes in the real world are not easily recognized in these models. Dynamic models or process models address precisely this issue. They emphasize changes that have Dynamic models taken place, are taking place or may take place sometime in the future. Dynamic models are inherently more complicated than static models, and usually require much more computation. Simulation models are an important class of dynamic models that allow the simulation of real world processes. Observe that our (El Nin~o) La Nina system can be called a static model as it stores state-of- affairs data such as the average December 1997 temperatures. But at the same time, it can also be considered a simple dynamic model, because it allows us to compare different states of affairs.
- In summary, modeling serves as a bridge between theory and the real world, providing a structured and systematic approach to representing, analyzing, and understanding complex systems and processes. It enhances our ability to make informed decisions, solve problems, and gain insights into the intricacies of the world around us.

4. Define Geographic field. Explain its different data type and values.

(Apr 19)

Ans:

- ~ A field is a geographic phenomenon that has a value 'everywhere' in the study area.
- **Data types and values :-**
 1. **Nominal data values**

Nominal data values are values that provide a name or identifier so that we can discriminate between different values, but that is about all we can do. Specifically, we cannot do true computations with these values. An example are the names of geological units. This kind of data value is called categorical data when the values assigned are sorted according to some set of non-overlapping categories. For example, we might identify the soil type of a given area to belong to a certain (pre-defined) category.
 2. **Ordinal data values**

Ordinal data values are data values that can be put in some natural sequence but that do not allow any other type of computation. Household income, for instance, could be classified as being either 'low', 'average' or 'high'. Clearly this is their natural sequence, but this is all we can say—we cannot say that a high income is twice as high as an average income.
 3. **Interval data values**

Interval data values are quantitative, in that they allow simple forms of computation like addition and subtraction. However, interval data has no arithmetic zero value, and does not support multiplication or division. For instance, a temperature of 20 °C is not twice as warm as 10 °C, and thus centigrade temperatures are interval data values, not ratio data values.
 4. **Ratio data values**

Ratio data values allow most, if not all, forms of arithmetic computation. Rational data have a natural zero value, and multiplication and division of values are possible operators (distances measured in metres are an example). Continuous fields can be expected to have ratio data values, and hence we can interpolate them.

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5. Write a note on Topology and spatial relationships.**(Apr 19)****Ans:**

- ~ Topology deals with spatial properties that do not change under certain transformations. For example, features drawn on a sheet of rubber can be made to change in shape and size by stretching and pulling the sheet. However, some properties of these features do not change:
- Area E is still inside area D,
- The neighborhood relationships between A,B,C,D, and E stay intact, and their boundaries have the same start and end nodes, and
- The areas are still bounded by the same boundaries, only the shapes and lengths of their perimeters have changed.

Topological relationships are built from simple elements into more complex elements: nodes define line segments, and line segments connect to define lines, which in turn define polygons.

Topological relationships

The mathematical properties of the geometric space used for spatial data can be described as follows:

- The space is a three-dimensional Euclidean space where for every point we can determine its three-dimensional coordinates as a triple (x,y,z) of real numbers. In this space, we can define features like points, lines, polygons, and volumes as geometric primitives of the respective dimension. A point is zero-dimensional, a line one-dimensional, a polygon two-dimensional, and a volume is a three-dimensional primitive.
- The space is a metric space, which means that we can always compute the distance between two points according to a given distance function. Such a function is also known as a metric.
- The space is a topological space, of which the definition is a bit complicated. In essence, for every point in the space we can find a neighbourhood around it that fully belongs to that space as well.
- Interior and boundary are properties of spatial features that remain invariant under topological mappings. This means that under any topological mapping, the interior and the boundary of a feature remains unbroken and intact.

6. Explain the temporal dimension using suitable example.**(Apr 19)****Ans:**

- ❖ Geographic phenomena are also dynamic; they change over time.
- **Examples of the kinds of questions involving time include:**
 - Where and when did something happen?
 - How fast did this change occur?
 - In which order did the changes happen?
- **Representing time in GIS:**
 - Spatiotemporal data models are ways of organizing representations of space and time in a GIS.
 - The most common of these is a 'snapshot' state that represents a single point in time of an ongoing natural or man-made process.
 - We may store a series of these snapshot states to represent change.
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- **Different 'concepts' of time:**
 - Discrete and continuous time: Time can be measured along a discrete or continuous scale.
 - Discrete time is composed of discrete elements (seconds, minutes, hours, days, months, or years).
 - In continuous time, no such discrete elements exist, and for any two different points in time, there is always another point in between. Derive temporal relationships between events and periods such as 'before', 'overlap', and 'after'.
 - Valid time and transaction time: Valid time (or world time) is the time when an event really happened, or a string of events took place. Transaction time (or database time) is the time when the event was stored in the database or GIS.
 - Linear, branching and cyclic time: Time can be linear, extending from the past to the present ('now'), and into the future. Branching time—in which different time lines from a certain point in time onwards are possible—and cyclic time—in which

repeating cycles such as seasons or days of a week are recognized.

- Time granularity: When measuring time, granularity is the precision of a time value in a GIS or database (e.g. year, month, day, second, etc.). Different applications may obviously require different granularity.
- Absolute and relative time: Time can be represented as absolute or relative. Absolute time marks a point on the time line where events happen (e.g. '6 July 1999 at 11:15 p.m.'). Relative time is indicated relative to other points in time (e.g. 'yesterday', 'last year', 'tomorrow', which are all relative to 'now', or 'two weeks later').
- Change detection: Studies of this type are usually based on some 'model of change', which includes knowledge and hypotheses of how change occurs for the specific phenomena being studied. It includes knowledge about speed of tree growth.
- Spatiotemporal analysis: we consider changes of spatial and thematic attributes over time.
- We can keep the spatial domain fixed and look only at the attribute changes over time for a given location in space.



7. What are Geospatial data, Geoinformation, quality and metadata? What are the key components of spatial data? Why do they play important role in assessment of data quality? (NOV 19)

Ans:

- **Geospatial Data:**

Geospatial data refers to information that has a geographic or spatial component, tying data to specific locations on the Earth's surface. It can include a wide range of information such as coordinates, addresses, or features like rivers, roads, and buildings. Geospatial data is crucial for various applications, including mapping, navigation, urban planning, environmental monitoring, and more.

- **Geoinformation:**

Geoinformation is a broader term that encompasses not only geospatial data but also the processes of collecting, storing, analyzing, and presenting that data. It involves the use of geographic information systems (GIS) and other technologies to manipulate and derive meaningful insights from spatial data.

➤ **Quality and Metadata in Geospatial Data:**

- **Quality:**

Geospatial data quality refers to the accuracy, precision, completeness, and reliability of the spatial information. High-quality geospatial data ensures that decisions and analyses based on the data are trustworthy and accurate. Factors affecting data quality include data collection methods, equipment precision, and the currency of the information.

- **Metadata:**

Metadata provides essential information about geospatial data, describing its characteristics, source, quality, and other relevant details. It serves as documentation that helps users understand the content and context of the data. Metadata includes information about data provenance, accuracy, scale, coordinate reference systems, and any transformations applied to the data.

➤ **Key Components of Spatial Data:**

- **Geometry:**

Geometry represents the spatial location and shape of geographic features. It includes points, lines, and polygons that define the geographic entities being represented.

- **Attributes:**

Attributes are non-spatial information associated with geographic features. For example, a point on a map might have attributes such as population, temperature, or land use.

- **Coordinate Reference System (CRS):**

The CRS defines the spatial reference for the data, specifying how coordinates relate to locations on the Earth's surface. Common CRS include latitude and longitude or projected coordinate systems.

- **Topology:**

Topology describes the spatial relationships and connectivity between different geographic features. It includes information about adjacency, containment, and connectivity.

Importance of Spatial Data and Quality Assessment:

- **Decision-Making:**

Many decisions in areas such as urban planning, emergency response, and natural resource management rely on accurate and reliable geospatial data. Assessing and ensuring data quality is crucial for making informed decisions.

- **Risk Management:**

Poor-quality spatial data can lead to errors in analyses and decision-making, potentially resulting in significant consequences. Assessing data quality helps identify and mitigate risks associated with inaccurate or unreliable information.

- **Interoperability:**

Spatial data from different sources often need to be integrated for comprehensive analyses. Assessing data quality and having standardized metadata ensure interoperability and compatibility between diverse datasets.

- **Public Trust:**

In applications where geospatial data is used for public services or communication, maintaining the trust of the public is essential. High-quality data and transparent metadata contribute to public confidence in the accuracy and reliability of spatial information.

- **Resource Management:**

Efficient and sustainable management of natural resources, land use, and infrastructure development relies on accurate geospatial data. Assessing data quality is essential for effective resource planning and management.

In conclusion, geospatial data, geoinformation, quality, and metadata are integral components in the realm of spatial information. They play a vital role in ensuring the accuracy, reliability, and usefulness of geographic data for a wide range of applications, contributing to informed decision-making and effective resource management

8. Explain the concept of Spatialtemporal data models. Explain the different concepts of time. (NOV19)

ANS:

Spatial-temporal data models are designed to represent and analyze information that varies both in space and time. These models are essential for applications where understanding the evolution of phenomena over both space and time is critical. Spatial-temporal data can be found in various fields, including meteorology, environmental science, transportation, epidemiology, and many others.

These models typically extend traditional spatial data models by incorporating the temporal dimension, allowing for the representation of how data changes over time. There are several key components and concepts associated with spatial-temporal data models.

- **Concepts of Time:-**

When dealing with spatial-temporal data models, various concepts of time are important to consider. Here are different temporal concepts:

- **Instant:**

An instant is a specific point in time, representing an exact moment. It is often associated with timestamps that provide a reference to when an event or observation occurred.

- **Interval:**

An interval represents a duration of time, indicating the time span between two instants. Intervals are used to describe the duration of events or changes.

- **Duration:**

Duration refers to the length of time that an event or state persists. It is the temporal extent between the start and end of an interval.

- **Time Series:**

A time series is a sequence of data points ordered in time. Each data point corresponds to a specific instant, and the collection of points forms a temporal record.

- **Temporal Resolution:**

Temporal resolution refers to the level of detail or granularity in representing time. It can vary from high-resolution data, capturing events at fine time scales, to low-resolution data, representing events at broader time intervals.

- **Temporal Relationships:**

Temporal relationships describe the order or sequence of events in time. Common relationships include "before," "after," "simultaneous," and "concurrent."

Understanding and modeling these temporal concepts are crucial for accurately representing and analyzing dynamic processes in spatial-temporal data models.

Whether it's tracking the movement of objects, analyzing environmental changes, or studying the spread of diseases, incorporating a temporal dimension enhances the comprehensiveness and utility of spatial data models.

9. Define Geographic Objects. Explain four parameters that define it.

(NOV 19)

Ans:

Geographic objects are entities or features in the real world that have a spatial location and can be represented in a geographic information system (GIS) or spatial database. These objects can range from simple point locations to complex polygons representing areas or volumes. Geographic objects are fundamental to spatial data modeling and are used to represent various real-world phenomena in a spatial context.

➤ **Four Parameters that Define Geographic Objects:**

- **Geometry:**

The geometry of a geographic object refers to its spatial shape or location on the Earth's surface. The geometry can be represented as points, lines, or polygons. For example:

Point: Representing a specific location with a single pair of coordinates (latitude and longitude).

Line: Representing a path or a linear feature, connecting multiple points.

Polygon: Representing a bounded area with a closed loop of connected points.

- **Attributes:**

Attributes are non-spatial properties or characteristics associated with a geographic object. These attributes provide additional information about the object, such as its name, population, temperature, or any other relevant data. For example, a point representing a city may have attributes like population size and elevation.

- **Spatial Reference System:**

The spatial reference system (SRS) defines how geographic objects are located and oriented in space. It includes information about coordinate reference systems, units of measurement, and the Earth's shape. Common SRS include latitude and longitude or projected coordinate systems.

- **Topology:**

- Topology describes the spatial relationships and connectivity between different geographic objects. It includes information about how objects are related in terms of adjacency, containment, connectivity, and other spatial relationships. For example:

Adjacency: Whether two objects share a common boundary.

Containment: Whether one object is completely within another.

Connectivity: Whether objects are connected or adjacent to each other.

These four parameters collectively define the essential characteristics of geographic objects and form the basis for spatial data modeling. When modeling geographic data, it's crucial to consider both the spatial and attribute components, ensuring accurate representation and analysis of real-world phenomena. Geographic objects, along with their geometry, attributes, spatial reference, and topology, enable the creation of detailed and comprehensive spatial databases used for mapping, analysis, and decision-making in various fields.

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10. Write a note on Irregular Tessellations.

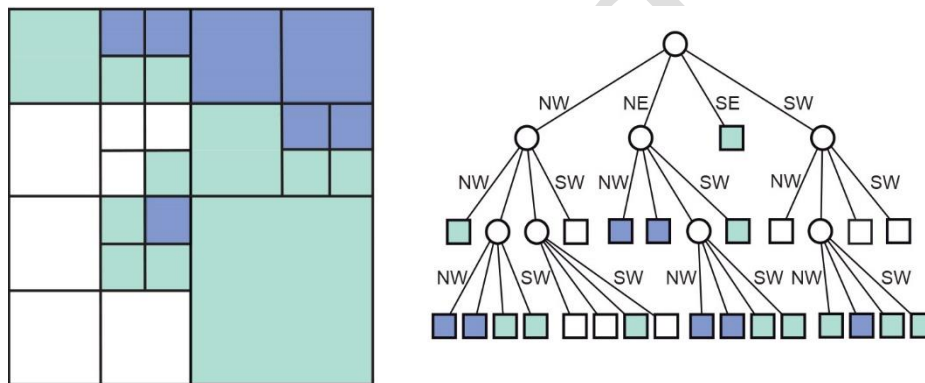
(NOV 19)

Ans:

❖ Irregular tessellations:

These are partitions of space into mutually disjoint cells, but now the cells may vary in size and shape, allowing them to adapt to the spatial phenomena that they represent. A well-known data structure in this family—upon which many more variations have been based—is the region quad tree. It is based on a regular tessellation of square cells but takes advantage of cases where neighbouring cells have the same field value, so that they can together be represented as one bigger cell.

A simple illustration is provided in Figure. It shows a 8×8 raster with three possible field values: white, green and blue. The quad tree that represents this raster is constructed by repeatedly splitting up the area into four quadrants, which are called NW, NE, SE, SW for obvious reasons. This procedure stops when all the cells in a quadrant have the same field value. The procedure produces an upside down, tree-like structure, known as a quad tree.



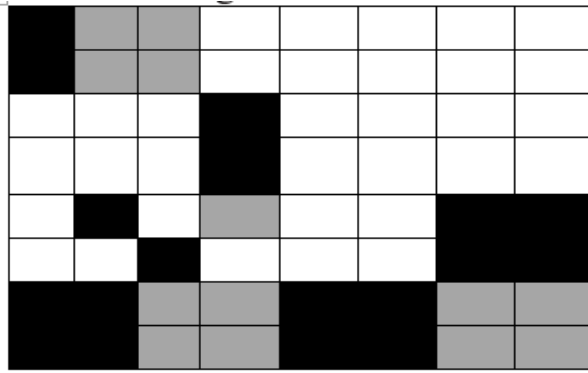
In conclusion, irregular tessellations in GIS offer a valuable alternative for representing and analyzing geographic features with irregular shapes and complex boundaries. The flexibility and adaptability of irregular tessellations make them well-suited for applications that require a more nuanced representation of real-world spatial phenomena.

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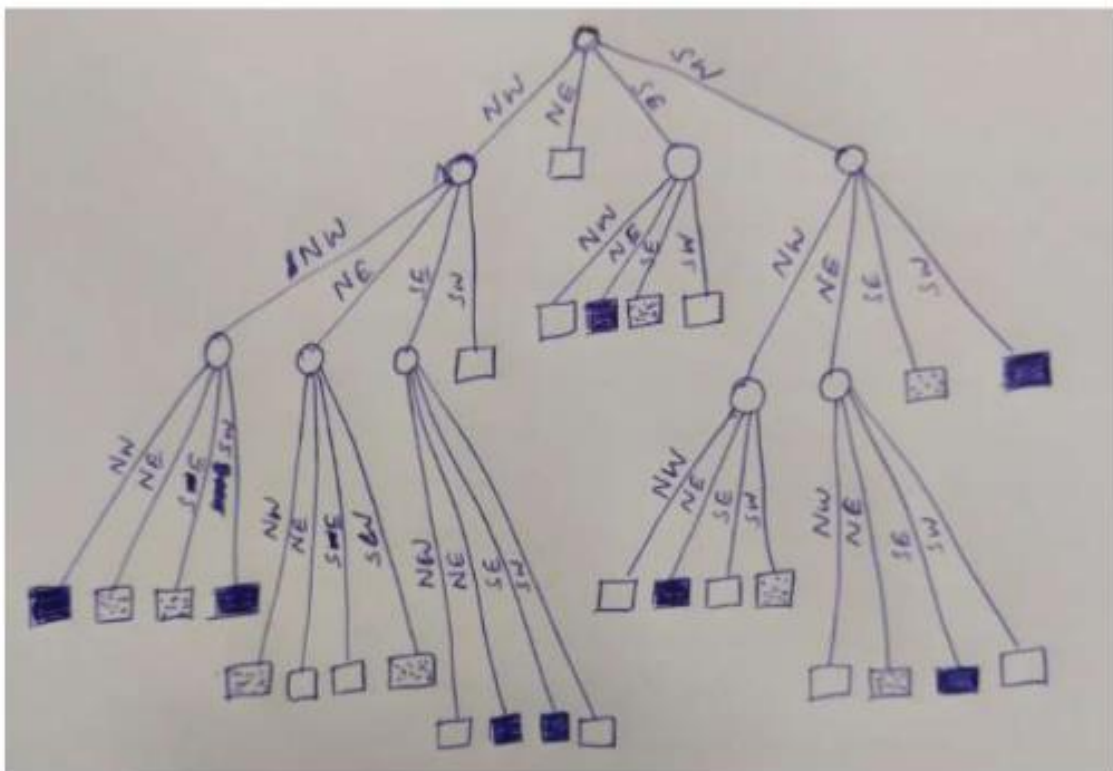
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11. Construct a quad tree for the following three valued raster. (NOV 19)



ANS:-



12. What is a Spatial Data and Spatial Analysis? Explain using suitable example

(APR 23)

Ans:

❖ Spatial Data:

Spatial data refers to any data that has a geographic or spatial component, meaning it is associated with a specific location on the Earth's surface. Spatial data can represent a wide range of features, including points, lines, polygons, and raster images. In Geographic Information Systems (GIS), spatial data is organized, stored, and analyzed to gain insights into spatial relationships, patterns, and trends.

❖ Spatial Analysis:

Spatial analysis in GIS involves the examination, manipulation, and interpretation of spatial data to extract meaningful patterns, relationships, and insights. It encompasses a variety of techniques and methods to analyze and understand the spatial components of data. Spatial analysis allows GIS professionals and researchers to make informed decisions, solve problems, and derive valuable information from geographic datasets.

❖ Example in GIS:

Let's consider an example involving spatial data and spatial analysis in a GIS context:

Scenario: Urban Planning and Land Use Analysis

❖ Spatial Data:

In this scenario, the spatial data might include various layers representing different aspects of the urban environment. These layers could include:

Land Parcels: Polygons representing individual plots of land.

Road Networks: Lines representing streets and highways.

Land Use Zoning: Polygons indicating different zones such as residential, commercial, or industrial.

Population Density: A raster layer representing the density of population across the urban area.

In summary, spatial data and spatial analysis in GIS provide a powerful framework for understanding and making decisions about the geographic aspects of various phenomena. In the context of urban planning, GIS enables professionals to analyze spatial data to make informed decisions regarding land use, infrastructure development, and community well-being.

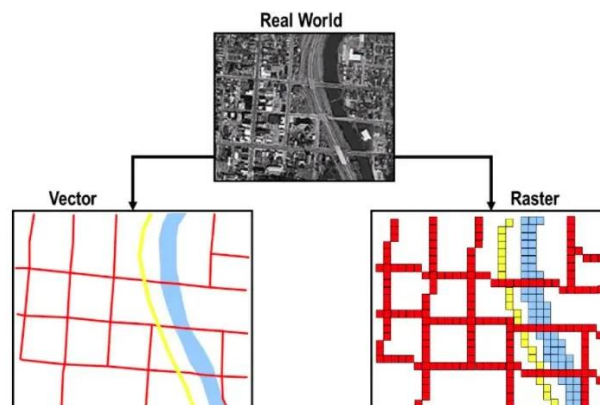
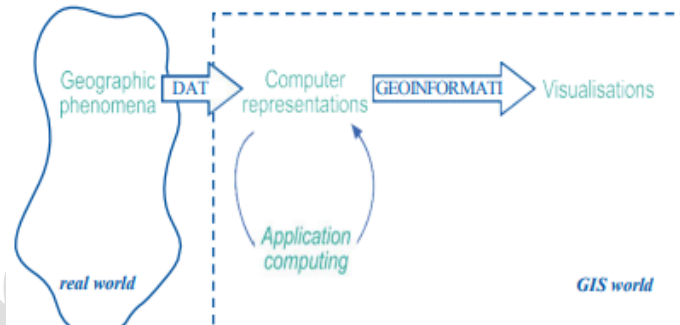
13. Define Model. Explain how models help in representing real world in GIS. (APR 23)

Ans:

GIS helps to analyse and understand more about processes and phenomena in the real world. Section 1.2.1 referred to the process of modelling, or building a representation which has certain characteristics in common with the real world. In practical terms, this refers to the process of representing key aspects of the real world digitally (inside a computer). These representations are made up of spatial data, stored in memory in the form of bits and bytes, on media such as the hard drive of a computer. This digital representation can then be subjected to various analytical functions (computations) in the GIS, and the output can be visualized in various ways.

Modelling is the process of producing an abstraction of the 'real world' so that some part of it can be more easily handled.

Depending on the application domain of the model, it may be necessary to manipulate the data with specific techniques. To investigate the geology of an area, we may be interested in obtaining a geological classification. This may result in additional computer representations, again stored in bits and bytes. To examine how the data is stored inside the GIS, one could look into the actual data files, but this information is largely meaningless to a normal user.



In order to better understand both our representation of the phenomena, and our eventual output from any analysis, we can use the GIS to create visualizations from the computer representation, either on-screen, printed on paper, or otherwise. It is crucial to understand the fundamental differences between these notions. The real world, after all, is a completely different domain than the 'GIS' world, in which we build models or simulations of the real world. The above two are types of representations of real world using vector and raster representation methods.

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14. Represent the given three valued raster using quad tree. (APR 23)

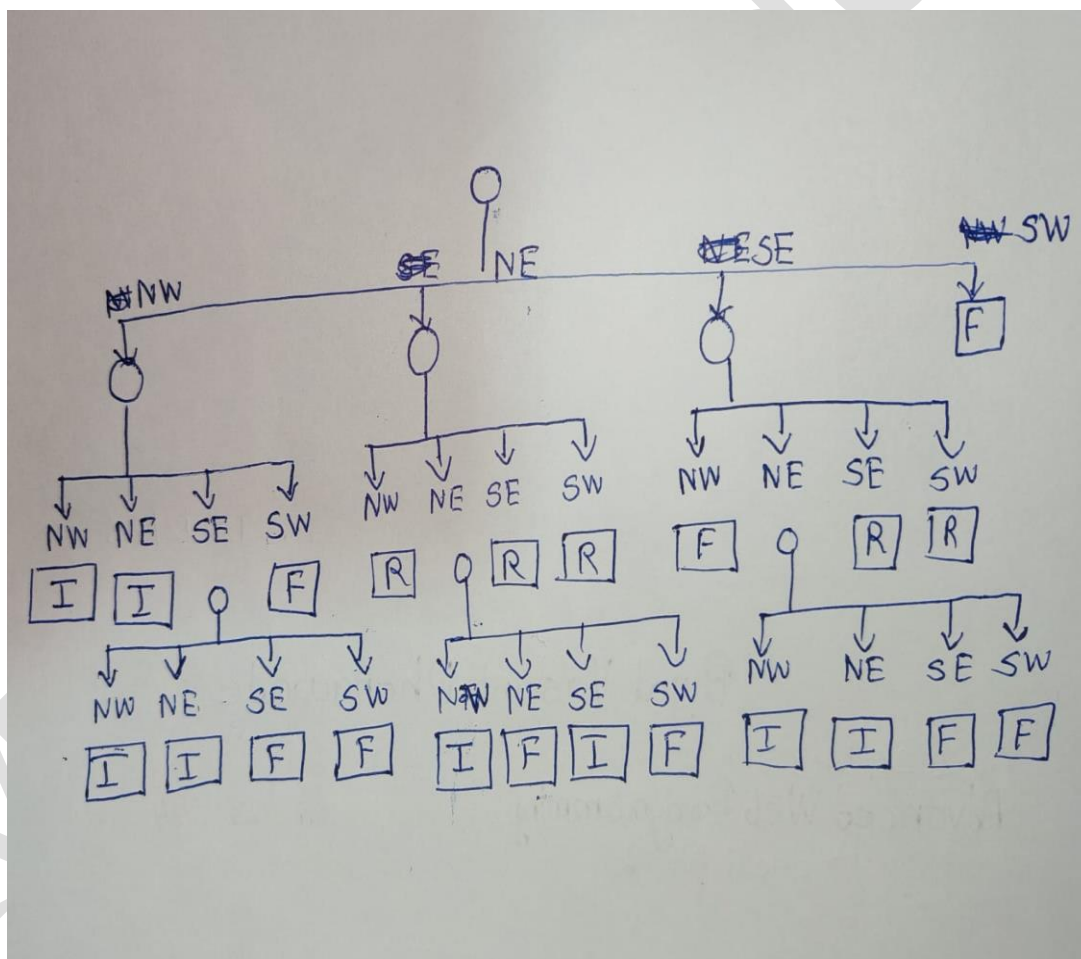
F- Forest LAN

I-Industrial Area

R- Residential Area

I	I	I	I	R	R	I	F
I	I	I	I	R	R	F	I
F	F	I	I	R	R	R	R
F	F	F	F	R	R	R	R
F	F	F	F	F	F	I	I
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F	F	F	F	R	R	R	R
F	F	F	F	R	R	R	R

ANS



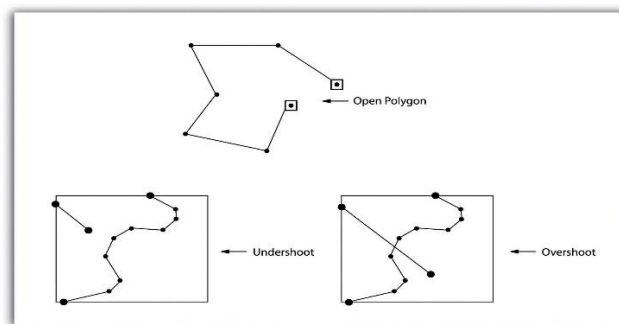
15. Explain the mathematical properties of geometric space used in spatial data using suitable diagram. (APR 23)

Ans:

- ❖ Geometric space in the context of GIS (Geographic Information Systems) refers to the mathematical representation of spatial features on the Earth's surface. It involves the use of geometric shapes and mathematical principles to model and analyze spatial data. Here are some key mathematical properties of geometric space used in GIS:-

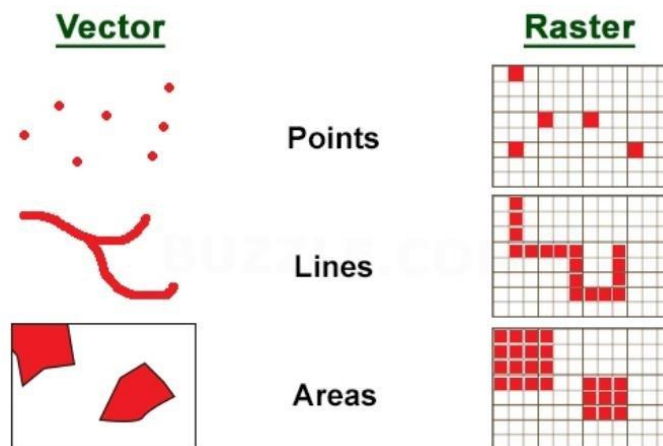
➤ **Topology:**

- Topology defines spatial relationships between geometric elements, such as adjacency, connectivity, and containment.
- Topological relationships include concepts like "contains," "intersects," and "touches."



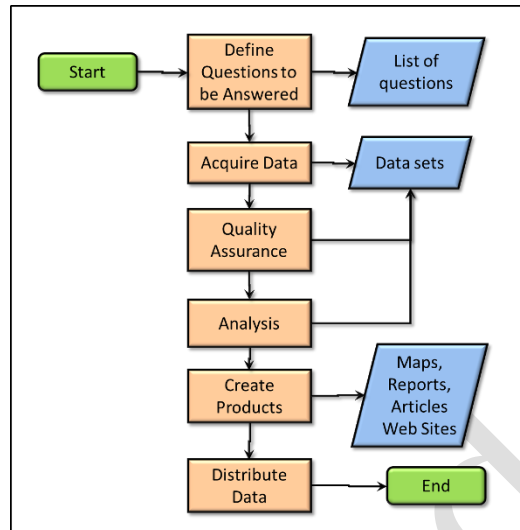
➤ **Vector and Raster Data:**

- **Vector Data**
 - Vector data in GIS represents geographic features as points, lines, and polygons.
 - Each feature has both spatial geometry and attribute information.
- **Raster Data**
 - Raster data in GIS is represented as a grid of cells or pixels, where each cell holds a value.
 - Suitable for continuous data representation.



➤ **Spatial Analysis:**

- Spatial analysis in GIS involves manipulating spatial data to derive meaningful insights.
- Operations include overlay analysis, buffering, proximity analysis, etc.



These mathematical properties are foundational in GIS, where spatial relationships and patterns are analyzed for decision-making in fields such as urban planning, environmental management, and resource allocation. GIS enables professionals to explore, visualize, and interpret spatial data to address complex challenges.



16. Spatiotemporal data model. Explain the concept of representing time in GIS. (APR23)

Ans:

Spatiotemporal data models are ways of organizing representations of space and time in a GIS. Several representations techniques have been proposed. Perhaps the most common of these is a 'snapshot' state that represents a single point in time of an ongoing natural or man-made process. We may store a series of these snapshot states to represent change, but must be aware that this is by no means a comprehensive representation of that process.

1. Discrete and continuous time

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4. Time granularity

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5. Absolute and relative time

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17. Write a short note on nature of GIS.**Ans:**

- A geographic information system (GIS) is a computer system for capturing, storing, checking, and displaying data related to positions on Earth's surface. By relating seemingly unrelated data, GIS can help individuals and organizations had better understand spatial patterns and relationships.
- **Following are the some of the examples of GIS where it is used:**
 - A biologist might be interested in the impact of slash-and-burn practices on the populations of amphibian species in the forests of a mountain range to obtain a better understanding of long-term threats to those populations;
 - A natural hazard analyst might like to identify the high-risk areas of annual monsoon-related flooding by investigating rainfall patterns and terrain characteristics;
 - A geological engineer might want to identify the best localities for construct- ing buildings in an earthquake-prone area by looking at rock formation characteristics;
 - A mining engineer could be interested in determining which prospective copper mines should be selected for future exploration, taking into account parameters such as extent, depth and quality of the ore body, among others;
 - A Geoinformatics engineer hired by a telecommunications company may want to determine the best sites for the company's relay stations, taking into ac- count various cost factors such as land prices, undulation of the terrain etc.
 - A forest manager might want to optimize timber production using data on soil and current tree stand distributions, in the presence of a number of operational constraints, such as the need to preserve species diversity in the area.
 - A hydrological engineer might want to study a number of water quality parameters of different sites in a freshwater lake to improve understanding of the current distribution of Typha reed beds, and why it differs from that of a decade ago.

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18. Write a short note on Geographic phenomenon.**Ans:**

- ❖ GIS operates under the assumption that the relevant spatial phenomena occur in a two- or three-dimensional Euclidean space, unless otherwise specified. Euclidean space can be informally defined as a model of space in which locations are represented by coordinates—(x, y) in 2D; (x, y, z) in 3D—and distance and direction can be defined with geometric formulas. In the 2D case, this is known as the Euclidean plane, which is the most common Euclidean space in GIS use. In order to be able to represent relevant aspects of real world phenomena inside a GIS, we first need to define what it is we are referring to. We might define a geographic phenomenon as a manifestation of an entity or process of interest that:
 - Can be named or described,
 - Can be georeferenced, and
 - Can be assigned a time (interval) at which it is/was present.

➤ Geographic Fields

A geographic field is a geographic phenomenon for which, for every point in the study area, a value can be determined. A field is a mathematical function f that associates a specific value with any position in the study area. Hence if (x, y) is a position in the study area, then $f(x, y)$ stands for the value of the field f at locality (x, y) .

➤ Geographic Objects

Geographic objects populate the study area, and are usually well-distinguished, discrete, and bounded entities. The space between them is potentially 'empty' or undetermined. When a geographic phenomenon is not present everywhere in the study area, but somehow 'sparsely' populates it, we look at it as a collection of geographic objects. Such objects are usually easily distinguished and named, and their position in space is determined by a combination of one or more of the parameters.

➤ Boundaries

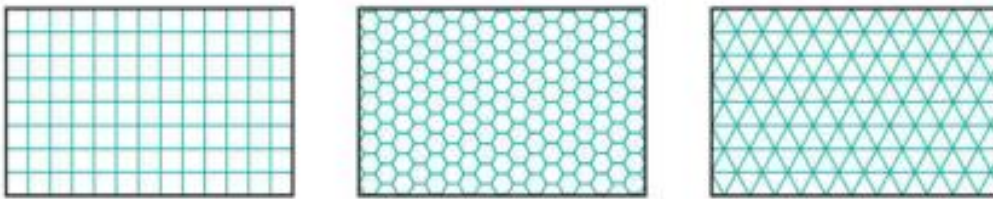
Where shape and/or size of contiguous areas matter, the notion of boundary is used. Boundary is used for geographic objects and for discrete geographic fields. Location, shape and size are fully determined if we know an area's boundary, so the boundary is a good candidate for representing it. Naturally crisp boundaries are one that can be determined with almost arbitrary precision, dependent only on the data acquisition technique applied. Fuzzy boundaries contrast with crisp boundaries in that the boundary is not a precise line, but rather itself an area of transition. Crisp boundaries are more common in man-made phenomena, whereas fuzzy boundaries are more common with natural phenomena.

19. What is regular and irregular tessellation? Explain it.

Ans:

1. Regular Tessellations:

A tessellation (or tiling) is a partitioning of space into mutually exclusive cells that together make up the complete study space. With each cell, some (thematic) value is associated to characterize that part of space. In a regular tessellation, the cells are the same shape and size. The simplest example is a rectangular raster of unit squares, represented in a computer in the 2D case as an array of $n \times m$ elements. Following are the three types of tessellations.



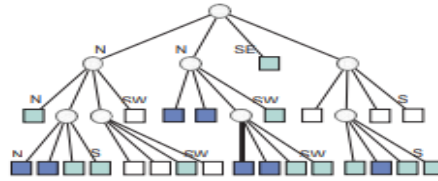
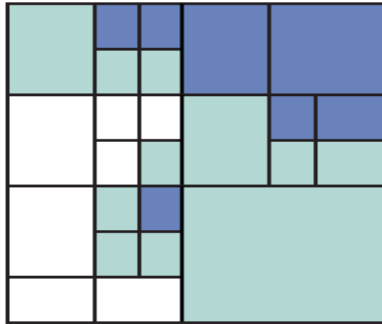
The three most common types of regular tessellation: from left to right, square cells, hexagonal cells and triangular cells.

In all regular tessellations, the cells are of the same shape and size, and the field attribute value assigned to a cell is associated with the entire area occupied by the cell. The square cell tessellation is by far the most used, mainly because georeferencing a cell is so straightforward. These tessellations are known under various names in different GIS packages, but most frequently as rasters.

2. Irregular Tessellations:

Irregular tessellations are more complex than the regular ones, but they are also more adaptive, which typically leads to a reduction in the amount of memory used to store the data. A well-known data structure in this family—upon which many more variations have been based—is the region quadtree. It is based on a regular tessellation of square cells but takes advantage of cases where neighbouring cells have the same field value, so that they can together be represented as one bigger cell. A simple illustration is provided in Figure.

It shows a small 8×8 raster with three possible field values: white, green and blue. The quadtree that represents this raster is constructed by repeatedly splitting up the area into four quadrants, which are called NW, NE, SE, SW for obvious reasons. This procedure stops when all the cells in a quadrant have the same field value. The procedure produces an upside-down, tree-like structure, known as a quadtree. In main memory, the nodes of a quadtree (both circles and squares in the figure below) are represented as records. The links between them are pointers, a programming technique to address (i.e. to point to) other records.



20. What is topology and spatial representations of geographic objects.

Ans:

- ❖ Topology deals with spatial properties that do not change under certain transformations. For example, features drawn on a sheet of rubber can be made to change in shape and size by stretching and pulling the sheet. However, some properties of these features do not change:
 - Area E is still inside area D,
 - The neighbourhood relationships between A,B,C,D, and E stay intact, and their boundaries have the same start and end nodes, and
 - The areas are still bounded by the same boundaries, only the shapes and lengths of their perimeters have changed.

Topological relationships are built from simple elements into more complex elements: nodes define line segments, and line segments connect to define lines, which in turn define polygons.

Topological relationships

The mathematical properties of the geometric space used for spatial data can be described as follows:

- The space is a three-dimensional Euclidean space where for every point we can determine its three-dimensional coordinates as a triple (x,y,z) of real numbers. In this space, we can define features like points, lines, polygons, and volumes as geometric primitives of the respective dimension. A point is zero-dimensional, a line one-dimensional, a polygon two-dimensional, and a volume is a three-dimensional primitive.
- The space is a metric space, which means that we can always compute the distance between two points according to a given distance function. Such a function is also known as a metric.
- The space is a topological space, of which the definition is a bit complicated. In essence, for every point in the space we can find a neighbourhood around it that fully belongs to that space as well.

Interior and boundary are properties of spatial features that remain invariant under topological mappings. This means that under any topological mapping, the interior and the boundary of a feature remains unbroken and intact.

21. Explain temporal dimension in brief with example.**Ans:**

- ❖ Besides having geometric, thematic, and topological properties, geographic phenomena are also dynamic; they change over time. For an increasing number of applications, these changes themselves are the key aspect of the phenomenon to study. Examples include identifying the owners of a land parcel in 1972, or how land cover in a certain area changed from native forest to pastures over a specific time. We can note that some features or phenomena change slowly, such as geological features, or as in the example of land cover given above. Other phenomena change very rapidly, such as the movement of people or atmospheric conditions. For different applications, different scales of measurement will apply.

Examples of the kinds of questions involving time include:

- Where and when did something happen?
- How fast did this change occur?
- In which order did the changes happen?

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